First-year medical students — Pr B. Boutabia-Chéraitia
Academic year : $2026 - 2027$

Problem Set 2 : Biophysics of solutions. Dialysis

Exercise 1:

Calculate the force required to burst a red blood cell, knowing that isotonicity corresponds to $9g/l$ of N_aCl, and complete hemolysis corresponds to $3g/l$ of N_aCl. The erythrocyte will be modeled as a cylinder of diameter $d = 7\mu m$ and thickness $e = 1\mu m$. Its membrane is assumed to be semi-permeable.

$Resp : 5, 29.10^5 Pa; 5, 23.10^{-5} N$.

Exercise 2:

An artificial kidney used in chronic hemodialysis includes a diffusion membrane (Cuprophan) with a surface area $S = 1m^2$. Across this membrane, the patient's blood on one side and a constantly renewed fluid (dialysate) - which is isotonic to plasma and free of urea - on the other side are brought into equilibrium. We will consider here only the problem of urea elimination. Recall that water and urea diffuse relatively rapidly across cell membranes.

1) Explain why:

a) the renewed fluid must be isotonic to plasma.

b) not only the blood but also all the water in the body, with a volume $V = 40l$, is thereby cleared of urea.

2)Towards what value should the blood urea concentration $c(t)$ tend after a very long period of hemodialysis? For what physiological reason does it not do so?

3) Assume that the urea concentration gradient is uniform across the thickness of the membrane, which has a permeability $P = 1.5 \times 10^{-6} m/s$ for urea. If the initial uremia at the start of the session is $C_0 = 2g/l$, calculate the initial mass flow rate of urea.

4) If the daily production of urea is $25g/day$, what will be the patient's uremia value after a very long time?

Exercise 3:

A patient arrives in the emergency department in a state of septic shock resulting in anuria. To compensate for the renal failure, it is decided to subject him to a peritoneal dialysis session. This procedure consists of injecting into the peritoneal cavity a volume V_e of a solution that is isotonic to plasma but free of urea.

Across the barrier formed by the capillary vessel walls and the peritoneal membrane, a diffusion equilibrium is established over a surface S between the blood circulating in the peritoneal vessels and the injected fluid.

1) Explain why there will be no change in the volume of the two internal and external compartments during dialysis.

2) If C_0 denotes the patient's initial uremia, what will the uremia C_∞ be when equilibrium is reached? The patient's body contains a volume $V_i = 40l$, and a volume $V_e = 2l$ is injected.

3) After a sufficiently long time to consider that equilibrium has been reached, the injected fluid is aspirated, then a second cycle is started, and so on. How many cycles will be required to reduce the patient's uremia from the initial value $C_0 = 1.2g/l$, to the near-normal value $C_n = 0.4g/l$?

4) The duration of each cycle is set to $\tau = 20\,\text{min}$. Assuming that diffusion equilibrium is reached by the end of this duration, calculate:

a) the total duration of dialysis for the patient's uremia to reach $0.4g/l$.

b) the urea clearance, that is, the virtual volume of blood completely cleared of its urea per unit of time, in ml/min.